

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 2 SET A

SUMMER SEMESTER, 2023-2024

DURATION: 30 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Use Karnaugh map to find the minimal form for the following expression. Clearly group the terms contributing to your solution.

5
(CO3)
(PO2)

$$F(A, B, C, D, E) = \sum m(0, 1, 4, 5, 8, 13, 16, 21, 22, 26, 29) + \sum d(6, 7, 17, 20, 25)$$

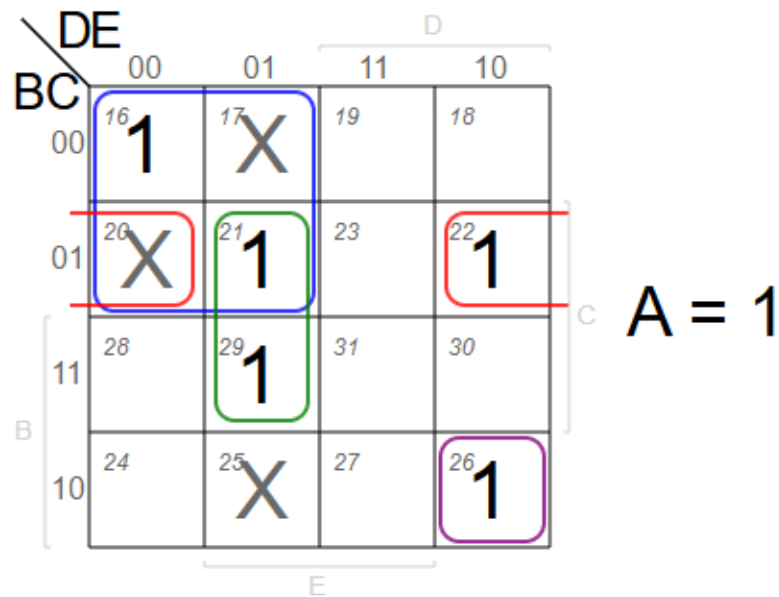
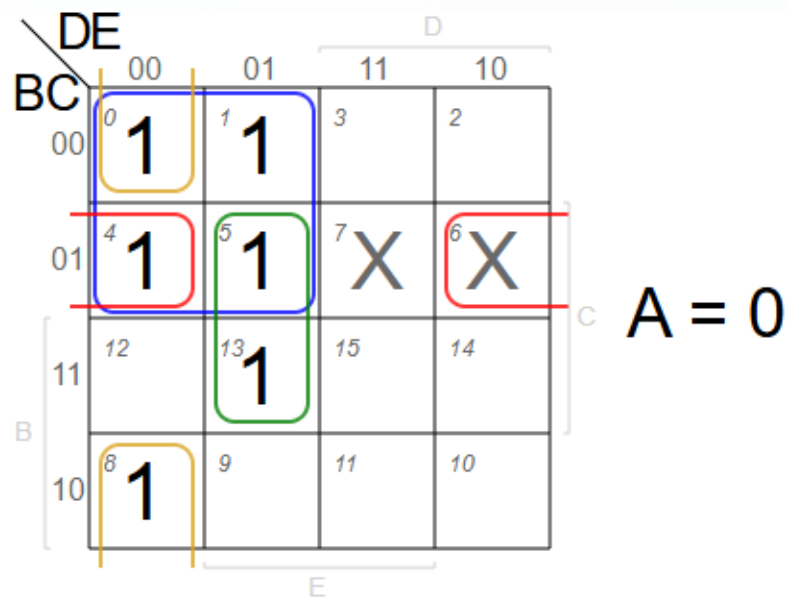
Solution:

Final Answer:

$$F = B'D' + B'CE' + CD'E + A'C'D'E' + ABC'DE'$$

Rubrics:

- 0.5 point each for correctly grouping each term (times 5).
- 0.5 point each for correctly identifying each term (times 5).
- No point for drawing the K-map since its already given.



$$F = B'D' + B'CE' + CD'E + A'C'D'E' + ABC'DE'$$

Figure 1: K-map solution

2. Simplify the following expression using Quine McCluskey (Tabular Method). And in the process identify the Prime Implicants and Essential Prime Implicants.

11
(CO5)
(PO3)

$$F(A, B, C, D) = \sum m(1, 2, 9, 11, 15) + \sum d(4, 13)$$

Solution:

For detailed solution visit this site.

Final Answer:

$$F = A'B'CD' + B'C'D + AD$$

Rubrics:

- 1 point for correctly writing the binary representation of each minterm.
- 1 point for correctly grouping minterms based on number of 1s in their binary representation.
- 3 points for correct matching table 1.
- 3 points for correct matching table 2.
- 2 points for correct Prime Implicant Table.
- 1 point for correctly identifying Essential Prime Implicants.

3. Simplify the following Boolean expression using the Cut and Try method, and draw the simplified Boolean circuit using only NOR gates.

4
(CO2)
(PO2)

$$F = ab'c + a'c + ab$$

Solution:

Given,

$$\begin{aligned} F &= a\bar{b}c + \bar{a}c + ab \\ \Rightarrow F &= a\bar{b}c + ab + \bar{a}c \\ \Rightarrow F &= a(\bar{b}c + b) + \bar{a}c \\ \Rightarrow F &= a(\bar{b} + b)(c + b) + \bar{a}c \\ \Rightarrow F &= a(c + b) + \bar{a}c \\ \Rightarrow F &= ac + ab + \bar{a}c \\ \Rightarrow F &= c(a + \bar{a}) + ab \\ F &= ab + c \end{aligned}$$

NOR Representation -

$$F = c + \bar{a} + \bar{b}$$

Try to draw it on your own.

Final Answer:

$$F = ab + c$$

$$F = c + \bar{a} + \bar{b}$$

Rubrics:

- 2 points for correct simplification.
- 2 points for correct NOR representation.